WHEN BETTER GETS WORSE: IMPROVEMENT FIDELITY OF SELF-IMPROVEMENT OPERATORS IN ADAPTIVE WORLDS

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Paper under double-blind review

ABSTRACT

Self-improvement systems repeatedly replace an incumbent policy with a candidate that a verifier says is better. The hidden assumption is that this verifier-approved improvement transfers after deployment, even when the deployment changes the world. This implication can fail even when the verifier correctly evaluates every individual policy.

We therefore define *Improvement Fidelity* over the update transition $\tau = (\pi_t, \pi_{t+1})$, rather than over isolated policy values. The proxy improvement is $\Delta_V = J_V(\pi') - J_V(\pi)$ and the deployment-induced improvement is $\Delta_* = \mathcal{J}(\pi') - \mathcal{J}(\pi)$. An *improvement reversal* is $\Delta_V > 0$ with $\Delta_* < 0$: a verifier-approved replacement that becomes worse after the world responds.

PIVOT (Paired Interventional Validation of Optimization Transitions) is a budgeted intervention mechanism that preserves update decisions. It uses paired rollouts and a differential correction $\Delta_* - \Delta_V$, then queries transitions whose high-fidelity result is most likely to change the selected update. It is not generic active learning and does not target global world-model error.

Across controlled mechanism and strategic fixture evidence, PIVOT exposes the transition-level failure while keeping the external stress test explicitly observational: the Binance audit finds no reversal (0/7 primary, 0/5 holdout). We do not claim causal market reversal, simulator ground truth, universal failure, or general multi-agent equilibrium; the contribution is a falsifiable evaluation criterion and a first budgeted implementation.

1 INTRODUCTION

Self-improvement is often described as a loop: generate a candidate, evaluate it, promote it, and repeat. But the central object optimized by the loop is not a policy in isolation; it is a *replacement operation*. A verifier may rank policies correctly while ranking improvements incorrectly, because deployment can change the transition and reward dynamics that define the next evaluation. The hidden implication is

$$\text{verifier says candidate is better} \implies \text{deployment policy is better,} \tag{1}$$

and this implication need not hold even for a fixed, external, correctly implemented verifier.

This failure is distinct from self-authored verification, ordinary dataset overfitting, or an incorrect evaluation program. It is a transfer failure for the learning update itself: the proxy world evaluates an object that changes when the candidate is deployed.

We study the local transition

$$\tau_t = (\pi_t, \pi_{t+1}) \qquad \text{rather than only} \qquad J(\pi_t). \tag{2}$$

Let V be a cheap observer or replay world and let $*$ denote a deployment world that may respond to the policy. The proxy and deployment improvements are $\Delta_V = J_V(\pi') - J_V(\pi)$ and $\Delta_* = \mathcal{J}(\pi') - \mathcal{J}(\pi)$. An improvement reversal is

$$\Delta_V > 0 \quad \wedge \quad \Delta_* < 0. \tag{3}$$

This definition makes no assumption about whether the proxy is learned, hand-coded, or externally audited. It asks whether the object consumed by the self-improvement loop transfers.

Our contributions are deliberately limited to four points:

1. **Contribution 1 — Failure mode.** We identify and formalize *improvement reversal*: a verifier-approved update can reduce deployed performance because deployment induces environmental response.
2. **Contribution 2 — Evaluation criterion.** We introduce *Improvement Fidelity* as the transfer criterion for self-improvement transitions and report IDE, ISC, IRR, SIRR, ISR, and CTI. The IMPROVE-X contract stores each transition and preserves unavailable world layers as explicit nulls.
3. **Contribution 3 — Method.** We introduce **PIVOT**, a paired differential evaluator and budgeted intervention rule that allocates queries to preserve update-selection decisions, not to minimize global prediction error.
4. **Contribution 4 — Evidence.** We provide controlled mechanism evidence, a strategic response fixture, and an external stress test with a checksum-bound observational finance audit. Nulls and open scientific gates remain explicit.

2 Related Work

Performative prediction formalizes the feedback from a deployed predictor to the data-generating process Perdomo et al. (2020). Performative RL and recent policy-gradient analyses extend this idea to policy-dependent transition and reward dynamics Basu et al. (2025); our focus is the local validity of a proposed update rather than optimization of a performatively stable policy. Policy-evaluation and world-model studies ask whether isolated values or rankings transfer from a learned world to a target environment Quevedo et al. (2025); Dann et al. (2026). We instead ask whether replacing one policy with another remains an improvement after deployment. This distinction matters because a simulator can rank policies well while misordering their local replacements.

Active learning typically selects labels to reduce global prediction error for a target y. PIVOT selects interventional rollouts to reduce the probability of a wrong update decision by estimating the paired difference $\Delta_* - \Delta_V$. It is therefore a decision-preservation procedure, not an unconditional value-model or generic uncertainty-sampling objective.

Self-evolving strategy systems and policy-evolution benchmarks make iterative improvement executable Mao et al. (2026); Wang et al. (2026), while self-authored verification work exposes failures when an agent controls its own tests Guo et al. (2026). PIVOT addresses a distinct failure mode: even a fixed external verifier can approve an update that is harmful after deployment-induced response. Multi-agent performative prediction and endogenous market simulators motivate the strategic layer Wang et al. (2025); Cheridito et al. (2025); they are testbeds here, not the paper's primary novelty.

3 Problem Formulation

Policy-induced worlds. Deployment can alter transitions, rewards, liquidity, or other agents' behavior. We write this response as $\mathcal{M}[\pi]$ and define the deployment value

$$\mathcal{J}(\pi) = J(\pi; \mathcal{M}[\pi]). \tag{4}$$

For exposition, three layers separate mechanisms rather than assert that they are always identifiable:

$$\Delta_{\text{direct}} = J(\pi'; \mathcal{M}[\pi]) - J(\pi; \mathcal{M}[\pi]), \tag{5}$$

$$\Delta_{\text{actor}} = J(\pi'; \mathcal{M}[\pi']) - J(\pi; \mathcal{M}[\pi]), \tag{6}$$

$$\Delta_{\text{strategic}} = J_i(\pi', BR_{-i}(\pi')) - J_i(\pi, BR_{-i}(\pi)). \tag{7}$$

The first ignores response, the second includes the focal agent's mechanical footprint, and the third includes adaptation by competitors. In experiments we report these as separate worlds and do not label a learned or observational proxy as ground truth.

Improvement fidelity metrics. For n transitions, IDE is $n^{-1}\sum_k |(\Delta_V)_k - (\Delta_*)_k|$. ISC is the fraction of non-tied transitions whose signs agree. IRR is $P(\Delta_* < 0 \mid \Delta_V > 0)$, while SIRR is $P(\Delta_{\text{strategic}} < 0 \mid \Delta_{\text{actor}} > 0)$. For a candidate set C_t, the update-selection regret is

$$\mathrm{ISR}_t = \max_{j \in C_t} (\Delta_*)_{t,j} - (\Delta_*)_{t,\hat{j}}, \tag{8}$$

where $\hat{j}$ is selected by the proxy or corrected estimate. CTI sums the true improvements of promoted updates over a trajectory. We call this metric the cumulative true improvement (CTI). All estimates below retain query counts, seeds, and confidence intervals in the artifact snapshot.

4 PIVOT

PIVOT stands for Paired Interventional Validation of Optimization Transitions. Given an incumbent and K candidates, it first computes cheap proxy deltas and pre-query features. The features include proxy delta, response strength, competition strength, candidate index, and an update footprint. The generic footprint records policy-distance and action/support shifts; the finance footprint records size, participation, urgency, holding, turnover, liquidity consumption, and spread crossing.

Paired differential correction. For a queried transition, incumbent and candidate are evaluated on the same initial state, exogenous path, seed, and opponent initialization. PIVOT fits

$$g_\theta(z_\Delta) \;\approx\; \Delta_* - \Delta_V, \qquad \widehat{\Delta}_* = \Delta_V + g_\theta(z_\Delta). \tag{9}$$

Pairing subtracts common-mode noise before aggregation. The unpaired ablation uses disjoint context seeds and the standard independent-sample standard error.

Active query rule. Let $\widehat{\Delta}_j$ and u_j be a corrected estimate and uncertainty. A transparent PIVOT score is

$$q_j = \frac{u_j(0.5 + r_j)(1 + o_j)}{c_j}, \tag{10}$$

where r_j is predicted sign risk, o_j is opportunity relative to the current best candidate, and c_j is high-fidelity cost. The rule is a decision-change heuristic, not a Bayesian optimality claim. Queried values replace predictions; unqueried candidates retain proxy/model estimates.

Why Transition Validation Differs from Active Learning. Traditional active learning asks which label will reduce a global prediction loss for y. PIVOT asks which paired intervention can change the ordering or sign of an update. Its target is the differential correction $\Delta_* - \Delta_V$, and its loss is update-selection regret rather than global world-model error. In short, **PIVOT is not designed to minimize global world-model error; it targets the smallest intervention budget required to preserve optimization decisions.** This distinction is operational: an uncertain transition whose high-fidelity value cannot change the selected candidate has lower priority than an otherwise similar transition near the selection margin.

5 THEORY

We state three modest results that motivate the differential estimand. They are not claims about a particular simulator architecture.

Proposition 1 (Global fidelity is sufficient)**.** *Suppose* $\sup_\pi |J_V(\pi) - J_*(\pi)| \le \varepsilon$. *Then every transition satisfies* $|\Delta_V - \Delta_*| \le 2\varepsilon$. *In particular, the sign is preserved whenever* $|\Delta_*| > 2\varepsilon$.

Proof. By adding and subtracting the two policy values, $\Delta_V - \Delta_* = [J_V(\pi') - J_*(\pi')] - [J_V(\pi) - J_*(\pi)]$. The triangle inequality gives the bound. If $|\Delta_*| > 2\varepsilon$, the perturbation cannot cross zero. □

Proposition 2 (Response–footprint bound)**.** *Let* $d(\pi, \pi')$ *be an update footprint and define the direct comparison in the incumbent response world,* $\Delta_{\text{direct}} = J(\pi'; \mathcal{M}[\pi]) - J(\pi; \mathcal{M}[\pi])$. *Assume the environment response satisfies* $D(\mathcal{M}[\pi'], \mathcal{M}[\pi]) \le L_M d(\pi, \pi')$ *and the value functional is* L_J*-Lipschitz in its world argument. Then*

$$|\Delta_{\text{actor}} - \Delta_{\text{direct}}| \le L_J L_M d(\pi, \pi'). \tag{11}$$

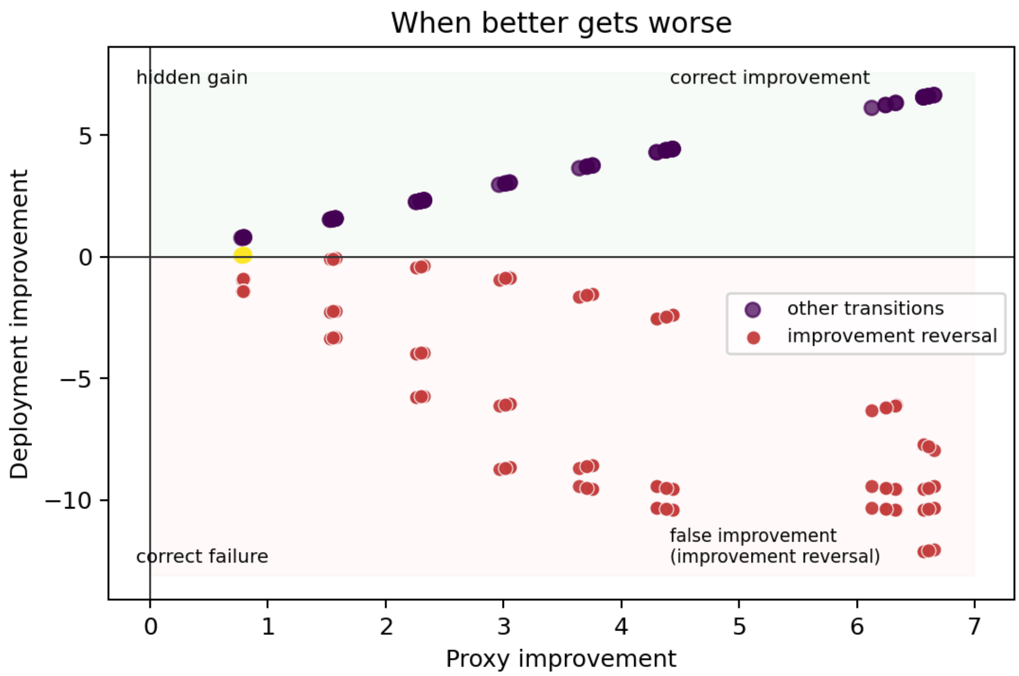

Figure 1: Controlled proxy versus deployment improvement. The upper-right region is correct improvement, the upper-left region is a hidden gain, the lower-left region is correct failure, and the lower-right region is the highlighted false improvement (improvement reversal). Each point is a transition in the frozen response–footprint grid.

Proof. The incumbent term cancels because both comparisons use $\mathcal{M}[\pi]$ as the reference world. The remaining candidate term is bounded by $L_J D(\mathcal{M}[\pi'], \mathcal{M}[\pi]) \leq L_J L_M d(\pi, \pi')$. The result is local; it does not assert that strategic best responses are globally Lipschitz. $\square$

Proposition 3 (Decision Preservation Under Differential Error)**.** *Consider two candidates i and j evaluated from the same incumbent, and suppose the true improvement margin is $m = \Delta_{*,i} - \Delta_{*,j} > 0$. If estimates satisfy $|\widehat{\Delta}_i - \Delta_{*,i}| < m/2$ and $|\widehat{\Delta}_j - \Delta_{*,j}| < m/2$, then $\widehat{\Delta}_i > \widehat{\Delta}_j$ and the selected update is unchanged.*

Proof. The estimated margin obeys $\widehat{\Delta}_i - \widehat{\Delta}_j = m + (\widehat{\Delta}_i - \Delta_{*,i}) - (\widehat{\Delta}_j - \Delta_{*,j})$. The two error terms have absolute sum strictly less than m, so the estimated margin remains positive. This is a local decision guarantee; it does not claim that PIVOT is globally optimal or that all candidates satisfy the margin condition. $\square$

The first proposition explains why global value accuracy is sufficient but unnecessarily strong: a policy-independent offset can be arbitrarily large without changing any improvement difference. The second motivates targeted fidelity: high footprint and high response sensitivity are the transitions most likely to change a decision. The third connects differential error to the selection decision that PIVOT must preserve. In strategic environments, small policy changes can still cause large competitor responses, so the bound should be treated as a diagnostic hypothesis rather than a universal smoothness assumption.

6 Controlled Experiments

Setup. The controlled world has finite horizon 12, bounded rewards, Gaussian exogenous noise, and a tunable response strength. Candidate policies change one intensity component. P2 uses three independent registered run groups and a response grid $\{0, 0.3, 0.7, 0.9\}$, footprint scales

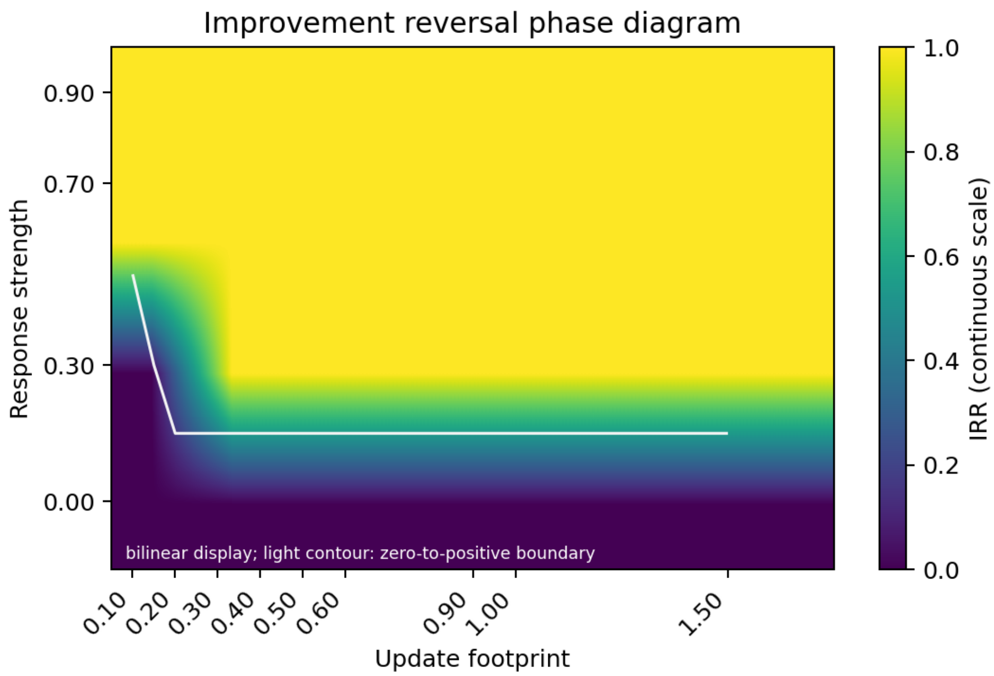

Figure 2: Continuous reversal-rate heatmap over response strength and update footprint. Bilinear display interpolation makes the registered grid's zero-to-positive reversal boundary visible; the structure is a controlled mechanism result, not a universal market law.

$\{0.1, 0.2, 0.3, 0.5\}$, and optimization strengths $\{0.5, 1, 1.5\}$. All primary comparisons use disjoint registered seeds and paired high-fidelity transitions.

Improvement reversal. Across 240 transitions per registered run, overall IRR is 0.7292 and the high-response IRR is 0.9722; low-response IRR is zero. The phase diagram in Figure 2 shows the intended response–footprint interaction. These results establish that a correct observer verifier can approve harmful updates in a world with policy-induced response.

Global versus differential fidelity. The matched-budget E4 comparison trains a global value model and a transition differential model on the same number of high-fidelity transition queries. Local-minus-global ISC is 0.1204; local-minus-global IDE is -1.7536. Update-selection regret is tied in this particular fixture, so we do not claim that a differential model dominates every policy-value baseline.

Value Fidelity versus Improvement Fidelity. We add a controlled evaluator contrast to isolate the estimand. Evaluator A has slightly lower isolated policy-value MAE (0.5489 versus 0.6000) and near-perfect policy ranking (0.9781 versus 1.0000), but its candidate-side bias grows with response and footprint. Evaluator B has a policy-independent value offset that is worse for isolated values but cancels in paired deltas. On the same 216 held-out transitions, A has IDE 1.0978, ISC 0.9444, IRR 0.1714, and ISR 0.2548, whereas B has IDE 0, ISC 1, IRR 0, ISR 0, and CTI -45.7229. A's CTI is -61.0081 (higher is better). This is a constructive controlled contrast, not a learned-model claim: similar policy-value accuracy does not determine local transition fidelity.

Budget frontier. At one high-fidelity query, PIVOT reduces ISR relative to Random by 4.7245 and relative to Top Proxy by 3.1301 in the registered fixture. Figure 4 shows the full frontier. The twelve-ablation suite retains a non-monotonic PIVOT curve across budgets; this is a robustness boundary, not a claim of monotonic benefit.

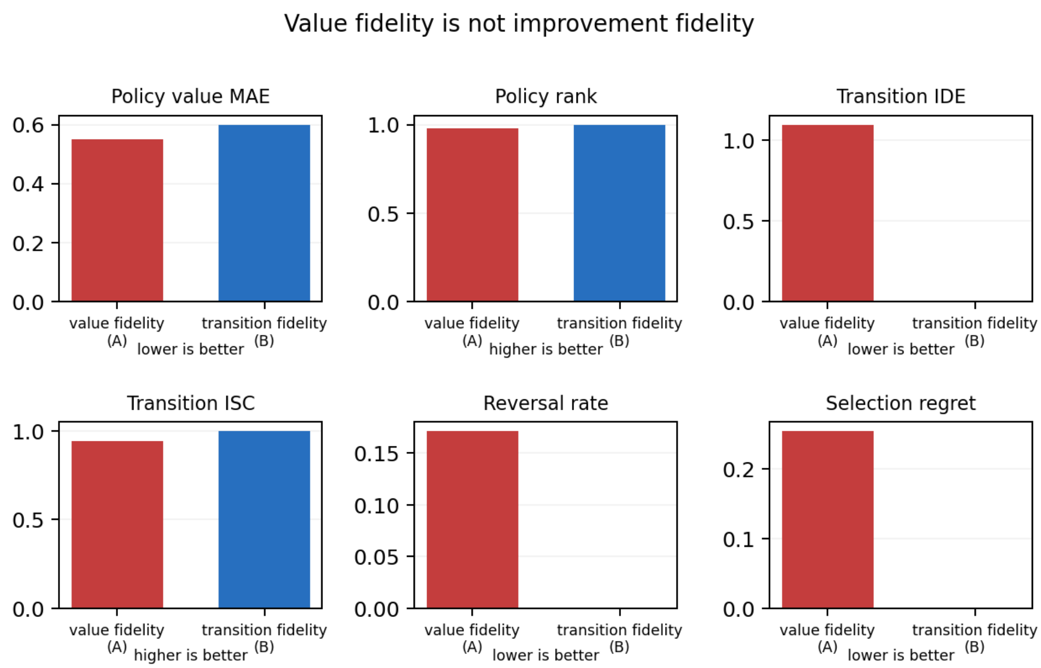

Figure 3: Policy-value and transition-differential predictions under matched high-fidelity budget. The estimands are related but not interchangeable.

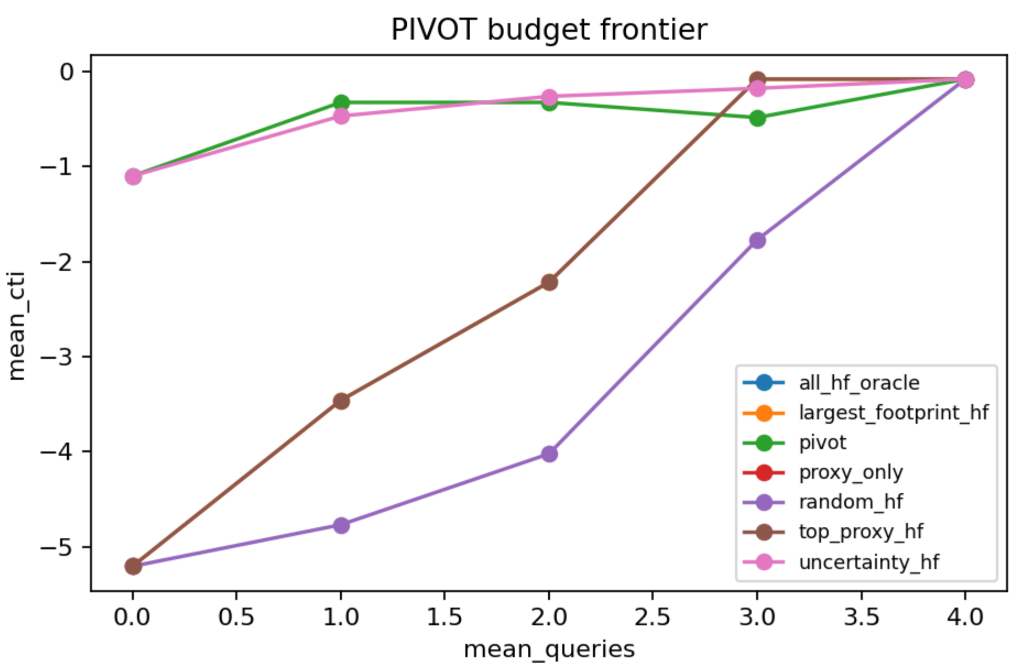

Figure 4: Matched-budget selection frontier. PIVOT spends queries on decision-sensitive transitions; All-HF is an oracle reference.

A required null. The E3 closed-loop fixture does not show proxy/true divergence: both curves increase together and reach the reward bound. We therefore report E3 as a negative stress test of the current fixture, not as evidence for "optimizing the wrong world." This distinction is important

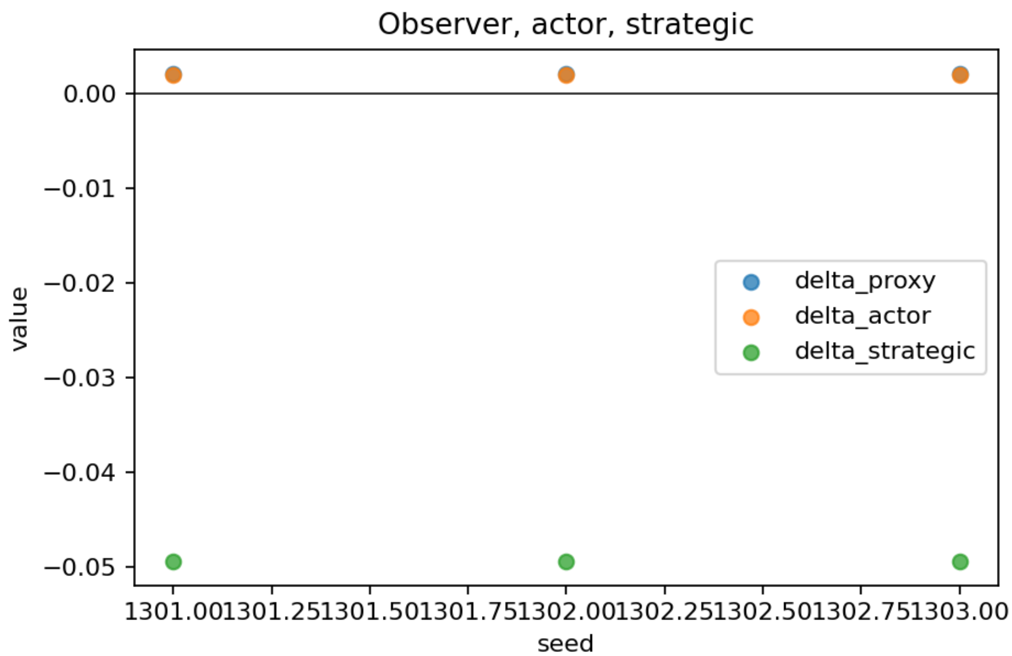

Figure 5: The same update evaluated in observer, actor, and strategic worlds.

because a paper can otherwise manufacture its central phenomenon by selecting a pathological environment after observing results.

7 Stress Tests Beyond Controlled Environments

Strategic response. Only the focal policy self-improves. Competitors are fixed, reactive, or finite-step adaptive. In the registered fixture, actor improvement is positive before adaptation, while adaptive competition produces SIRR 1.0 and competition effects of -0.0514 (E7) and -0.0405 (E8). The sensitivity contrast is -0.1082.

The opponent-mode sweep is reported in Appendix Figure 7. These results are mechanism evidence for the strategic layer, not a claim of general equilibrium or realistic market ecology. External strategic validation remains open.

Finance audit. The finance ladder is F0 historical path, F1 execution replay, F2 interactive virtual fills, and F4 strategic response. We additionally audit 12 sessions covering BTCUSDT, ETHUSDT, and BNBUSDT at four 2023 quarter starts using Binance's public one-minute klines and percentage book depth Binance (2023). The archive checksums are frozen in the project manifests. The depth world is an observational execution proxy: it cannot identify replenishment, hidden liquidity, post-trade response, or strategic adaptation.

At primary participation, seven sessions have positive F1 improvement and depth-proxy reversal is 0/7; the frozen holdout has 0/5 reversals. The pooled depth mechanical effect is -4.1554×10^{-7} with 95% bootstrap interval $[-5.3969, -2.9638] \times 10^{-7}$. Thus finance supplies a plausibility and boundary test, not causal confirmation of improvement reversal.

8 Ablations and Robustness

The frozen suite contains twelve required comparisons. Table 1 shows the compact main-text subset; the full aggregate, raw rows, and failure ledgers are included in the artifact snapshot.

Table 1: Selected controlled ablations. Values are descriptive estimates over three independent registered runs; slash-separated values compare variants.

Ablation	Estimate(s)	Metric
Paired vs unpaired	0.0433 / 0.0676	paired SE / unpaired SE
Footprint vs none	2.0668 / 2.0669	IDE; null
Small vs large	0.50 / 0.75	IRR
Weak vs strong response	0.458 / 1.000	IRR
Fixed vs adaptive	0.000 / 1.000	SIRR
Single vs multiple model	4.751 / 2.691	IDE
Candidate count 2 / 4	0.2368 / 0.2439	ISR
PIVOT budget 0 / 4	1.0210 / 0.0000	ISR

Pairing reduces standard error (0.0433 versus 0.0676). Multiple response levels improve held-out differential error relative to a single response level. However, footprint features are nearly tied with no-footprint features in this grid (IDE 2.0668 versus 2.0669), and candidate-count/HF-budget effects are small or non-monotonic. These nulls constrain the method claim: PIVOT is a budgeted transition-validation mechanism, not proof that every feature or budget increase helps.

9 Limitations and Claim Boundary

First, the controlled world is intentionally small and its response law is known. It establishes falsifiability of the estimand, not realism. Second, the public finance data are observational and use virtual fills; no live orders, broker connection, or private L2 data were used. Third, the strategic fixture is deterministic and requires validation in an independently calibrated competitive environment. Fourth, the current public typed update was selected exploratorily; a confirmatory update-generation and regime/holdout rule must be frozen before a new external claim. Fifth, E3's null means the current fixture does not demonstrate performative overoptimization. Finally, LLM/EvoQuant and M3/world-model adapters are intentionally deferred and are not needed for the core estimand.

The claim boundary is therefore precise: we validate learning progress at the policy-transition level in controlled adaptive worlds, demonstrate a strategic mechanism in a fixture, and report a negative observational finance audit. We do not certify actions, solve general multi-agent equilibrium, or claim that a learned simulator is ground truth.

10 Reproducibility and Conclusion

The paper snapshot contains 29 hash-indexed artifacts, seven figure/source-table pairs, the controlled evaluator-contrast rows, controlled gate summaries, the public expansion summary, and the full twelve-ablation aggregate. Three disjoint ablation runs reproduce with zero failed runs. The repository's full test suite and static checks are recorded in the release manifest. The IMPROVE-X transition contracts and controlled ImprovementBench smoke artifact are separately hash-bound in the repository. The build records the source commit and PDF verification metadata. No credentials or raw vendor archives enter the paper package.

The central lesson is simple: a self-improving agent should not be judged only by whether its new policy performs better in the world used to verify it. It should be judged by whether the update remains better in the world—and among the agents—that deployment causes to exist. PIVOT operationalizes this as paired, transition-level, budgeted interventional validation. The controlled evidence supports the need for that estimand; the external audits state exactly where evidence is still missing.

AI Use Statement

Generative AI tools were used by the authors during literature discovery, method and experiment design, implementation, mathematical drafting, prose editing, and figure-production support. The authors reviewed and tested all generated material, verified the reported computations, and take full responsibility for the paper, its claims, and its supplementary artifacts.

Reproducibility Statement

The anonymous supplement contains the source package, deterministic configuration files, hash-indexed snapshot, table/figure inputs, and commands needed to rebuild the reported artifact. The included public-data audit uses virtual fills only; no credentials, private data, raw vendor archives, or live orders are part of the submission package.

Ethics Statement

This work uses public market data and controlled virtual environments. It does not involve human subjects, personal data, live trading, or authorization of external actions.

References

Debabrota Basu, Udvas Das, Brahim Driss, and Uddalak Mukherjee. Performative policy gradient: Optimality in performative reinforcement learning, 2025. URL `https://arxiv.org/abs/2512.20576`.

Binance. Binance public data repository. `https://github.com/binance/binance-public-data`, 2023.

Patrick Cheridito, Jean-Loup Dupret, and Zhexin Wu. Abides-marl: A multi-agent reinforcement learning environment for endogenous price formation and execution in a limit order book, 2025. URL `https://arxiv.org/abs/2511.02016`.

Christoph Dann, Yishay Mansour, and Mehryar Mohri. Theoretical foundations and effective algorithms for policy-aware simulator learning, 2026. URL `https://arxiv.org/abs/2605.29032`.

Diandian Guo, Cong Cao, Fangfang Yuan, Yingqi Wang, Yueshan Wang, and Dakui Wang. Self-authored verification is unreliable in heuristic self-improving agents, 2026. URL `https://arxiv.org/abs/2607.24300`.

Jie Mao, Changlun Li, Xiang Li, Qiqi Duan, Jinhui Yuan, Xiang Liu, Yuyu Luo, Jing Tang, Xiaowen Chu, and Nan Tang. Evoquant: Self-evolving verifier-guided strategy optimization for robust quantitative trading, 2026. URL `https://arxiv.org/abs/2607.12455`.

Juan C. Perdomo, Tijana Zrnic, Celestine Mendler-Dunner, and Moritz Hardt. Performative prediction. *arXiv preprint arXiv:2002.06673*, 2020. URL `https://arxiv.org/abs/2002.06673`.

Julian Quevedo, Ansh Kumar Sharma, Yixiang Sun, Varad Suryavanshi, Percy Liang, and Sherry Yang. Worldgym: World model as an environment for policy evaluation, 2025. URL `https://arxiv.org/abs/2506.00613`.

Guanghui Wang, Krishna Acharya, Lokranjan Lakshmikanthan, Juba Ziani, and Vidya Muthukumar. Last-iterate convergence for symmetric, general-sum, 2×2 games under the exponential weights dynamic, 2025. URL `https://arxiv.org/abs/2502.08063`. Includes a multi-agent performative prediction application.

Zhilin Wang, Han Song, Runzhe Zhan, Jusen Du, Jiacheng Chen, Tianle Li, Qingyu Yin, Yulun Wu, Zhennan Shen, Tong Zhu, Yanshu Li, Guanjie Chen, Derek F. Wong, Yafu Li, Yu Cheng, and Yang Yang. Evopolicygym: Evaluating autonomous policy evolution in interactive environments, 2026. URL `https://arxiv.org/abs/2607.02440`.

A ARTIFACT AND FULL RESULT NOTES

The frozen artifact manifest is `../snapshot/manifest.json`. Its controlled summary hashes are inherited from the clean-room reproduction. The ablation aggregate is identified by the manifest hash prefix `e081f87f`; the public expansion remains marked `causal_impact_identified=false` and `live_orders=false`. Re-running the snapshot and table scripts from a new output directory is required before changing any result table.

Table 2: Controlled and public evidence summary.

Experiment	Estimate	Interpretation
P2 reversal	0.9722	IRR, high response
E4 local–global	0.1204	ISC difference
E5 Random–PIVOT	4.7245	ISR at HF budget 1
E5 Top Proxy–PIVOT	3.1301	ISR at HF budget 1
E6 zero participation	0.0000	F2–F1
E7 SIRR	1.0000	strategic reversal
E8 SIRR	1.0000	strategic reversal
E9 CTI	-1.5338	8-round exploratory loop

Table 3: Public expansion boundary.

Public audit field	Value
Asset/date sessions	12
F1-positive sessions	7
Primary reversals	0/7
Holdout reversals	0/5
Pooled depth effect	-4.1554×10^{-7}
Causal impact identified	No

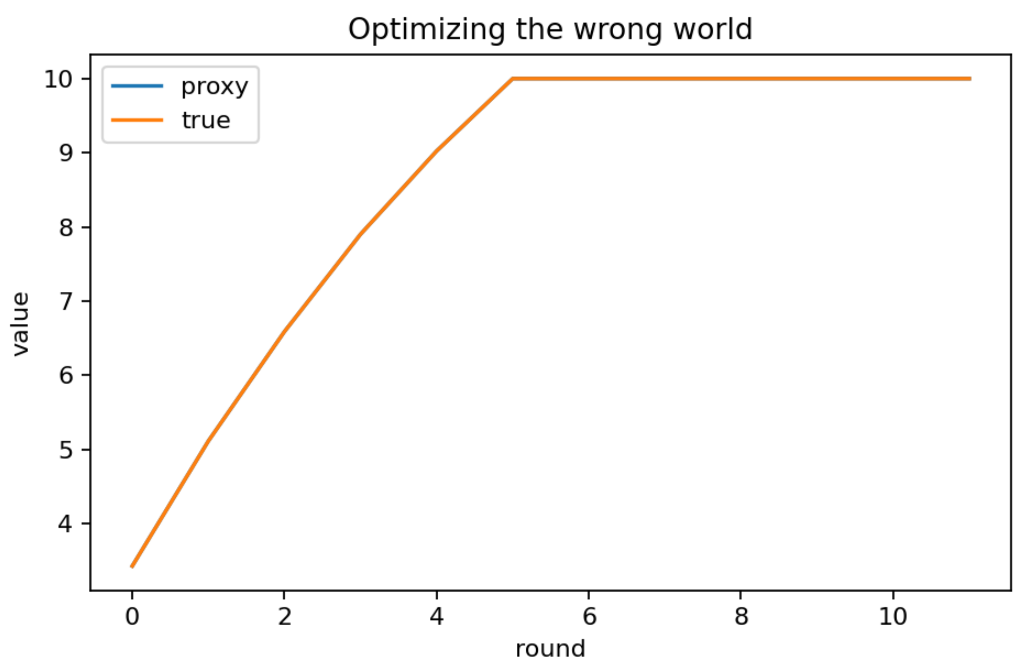

Figure 6: Required E3 null: proxy and true values overlap and saturate at the reward bound. The current closed-loop fixture therefore does not establish performative overoptimization.

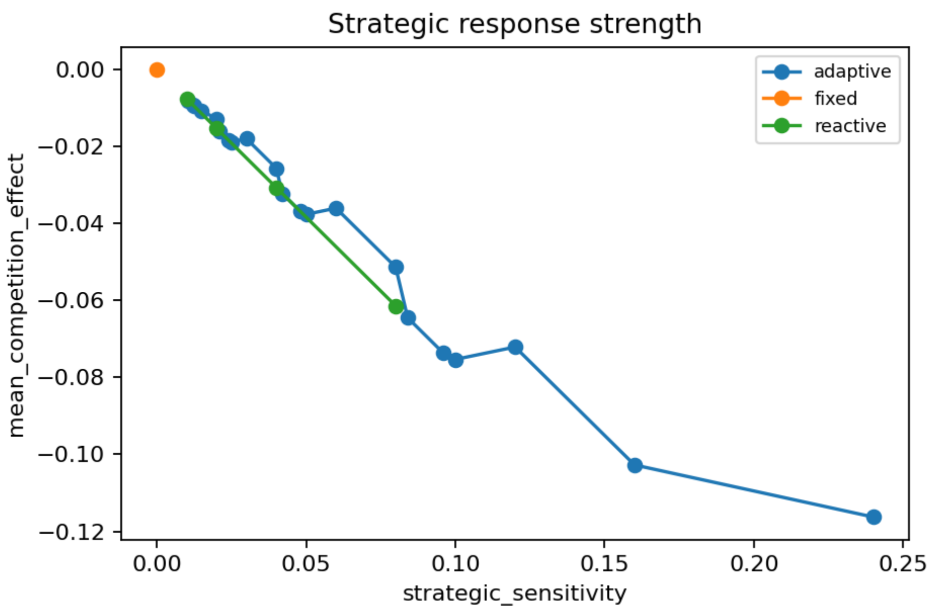

Figure 7: Strategic response effect across opponent modes and sensitivities. This appendix diagnostic is fixture evidence, not a general-equilibrium claim.